FI FISHINTEL GLOBAL

ENVIRONMENTAL RECONSTRUCTION REPORT

Decision-Time Environmental Reconstruction

Anchorage Failure During Forecast Storm Conditions

INCIDENT WINDOW	Winter storm period
REPORT REFERENCE	FIG-2025-001
VESSEL CATEGORY	Bulk carrier, in ballast
REPORT VERSION	v1.0
OPERATIONAL CONTEXT	UK coastal lee shore anchorage
CLASSIFICATION	Public — Anonymised

PIPELINE VERIFIED — RECONSTRUCTION DATA POPULATED

01

Executive Summary

Incident type: Grounding following anchor drag. A foreign-registered bulk carrier, anchored in ballast at a UK coastal anchorage, dragged its anchor during a severe winter storm and grounded on a public beach. Successfully refloated on the next high water. No injuries, no pollution.

Core question: What environmental conditions were objectively knowable at the time the decision was made to remain at anchor on a lee shore overnight, and at what point did those conditions make the vessel's position untenable?

Method: ERA5 atmospheric reanalysis data was extracted at the anchorage coordinates across a three-point time window (03:00–05:25 UTC) and processed through the FishIntel deterministic reconstruction pipeline. The reconstruction quantifies sustained wind, gust speed, significant wave height, atmospheric pressure, and visibility at each extraction point.

Key result: The vessel was in a HIGH-risk environmental regime before the anchor began to drag. ERA5 reconstruction shows sustained 10 m winds of between 25.3 and 27.0 knots (Beaufort force 6–7), gusts of 41.6–44.8 knots, and significant wave height of 3.8–4.1 metres across the incident window. The narrative description of "Force 9–10" appears to reflect the combined severity of locally generated wind waves and remotely generated swell propagating into the anchorage from the open English Channel — not sustained local wind at the 10 m reference height. The dominant forcing mechanisms for anchor drag were heavy swell loading on a vessel in ballast combined with severe gust intensity, rather than sustained storm-force local wind alone.

Analytical finding: Environmental reconstruction separates local wind forcing from remotely generated swell, revealing that the conventional Beaufort estimate in the narrative account conflated two distinct physical mechanisms.

The hazardous anchoring regime was already established by 03:00 UTC — more than 90 minutes before the anchor began to drag.

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Incident Overview

Vessel Profile

TYPE	Bulk carrier
FLAG	Foreign-registered
LENGTH OVERALL	~180 m
GROSS TONNAGE	~16,000
YEAR OF BUILD	Early 1980s
CARGO CONDITION	In ballast
PROPULSION	Direct drive, single fixed-pitch propeller, no bow thruster
CREW	18
ANCHORS	Two, AC14 type. Port: 11 shackles. Starboard: 10 shackles.
P&I INSURANCE	Not in force at time of incident

The vessel had sailed from a European port following a lengthy port state control detention. Unable to take bunkers before departure due to unpaid debts, she was ordered to proceed to North Africa for refuelling but had insufficient fuel for the passage. After several days alternating between anchoring and loitering at slow speed off the south-west coast of England, the vessel was authorised to take bunkers at a local port.

She anchored in the bay approximately 1 nautical mile off a public beach, in a charted depth of 17 metres, using the port anchor with 7 shackles of cable

(scope ratio approximately 9:1 at high water). A delivery of marine diesel was completed, but a subsequent lube oil delivery was not fulfilled. The vessel was unable to embark on her intended passage.

On the evening before the grounding, the master was aware that the weather forecast predicted southerly winds up to Beaufort force 8. He wrote in his night orders that the main engine should be at immediate readiness for wind speeds exceeding 16–17 m/s (31–33 knots). He discussed the risk of the vessel dragging anchor with the bridge watchkeepers.

By 0400 the following morning, the wind was gusting over 52 knots with a 4–5 metre swell. At 0425, the anchor began to drag. The master was called, began heaving the anchor, and attempted to proceed to sea. The anchor appeared to be fouled by a substantial length of chain, which distracted the crew and cost critical time. The vessel fell beam-to the wind, making over 2 knots leeway towards the shore. Despite full engine power and helm, the master was unable to turn the vessel into the wind. She grounded at 0525.

Local harbour resources coordinated the refloating operation. With the weather moderating, the vessel was pulled off the beach by harbour tugs at 1415 the same day, assisted by her own main engine. An underwater survey subsequently revealed hull plate deformation, breached tanks, and partial bilge keel detachment. There were no injuries and no pollution.

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Environmental Escalation Timeline

ERA5 reanalysis data was extracted at the anchorage coordinates across three timestamps spanning the critical period from storm escalation through anchor drag to grounding. The following table presents the reconstructed conditions at each extraction point.

TIMESTAMP (UTC)	EVENT	SUSTAINED WIND	GUSTS	SIG. WAVE HEIGHT	RISK REGIME
03:00	*Storm escalation*	25.3 kt	41.6 kt	3.8 m	HIGH
04:00	*Vessel yawing / anchor drag imminent*	26.2 kt	43.7 kt	4.0 m	HIGH
05:25	*Grounding*	27.0 kt	44.8 kt	4.14 m	HIGH

INTERPRETATION

The vessel did not enter a dangerous regime at the moment of grounding. The hazardous environmental state already existed at 03:00 UTC — more than 90 minutes before the anchor began to drag at 04:25. Conditions escalated continuously but moderately across the three-hour window: sustained wind increased by 1.7 knots, gusts by 3.2 knots, and significant wave height by 0.3 metres.

This profile is consistent with a deepening synoptic-scale storm system, not a sudden localised squall. The conditions that caused the anchor to drag were foreseeable from the environmental state that existed well before the event.

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Narrative Weather Description

The following environmental conditions were described in the original accident investigation narrative. These represent the qualitative and semi-quantitative descriptions available in the published account.

PARAMETER	NARRATIVE DESCRIPTION	OBSERVATIONS
Wind (sustained)	Southerly, Beaufort force 9–10 Equivalent to 47–55 knots	*Described at time of incident. No hourly progression provided.*
Wind (gusts)	"gusting over 52 knots" Reported by 0400	*Single threshold. No peak gust value stated.*
Wind direction	Southerly	*Cardinal direction only. No bearing given.*
Sea / swell	4–5 m swell	*Range given. No wave period or direction stated.*
Forecast (evening before)	Southerly up to Beaufort force 8 34–40 knots	*Forecast available to master. Conditions exceeded this forecast.*
Atmospheric pressure	Not stated	*No barometric data in narrative account.*
Air temperature	Not stated	—
Visibility	Not stated	—

Analytical note: The narrative account provides a description of conditions at the time of the incident but does not quantify the environmental progression in the hours before the anchor dragged. Several parameters relevant to the decision environment — atmospheric pressure, temperature, visibility — are absent from the narrative.

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Environmental Reconstruction

The following table compares the environmental conditions described in the narrative account with the values reconstructed by FishIntel using ECMWF ERA5 atmospheric reanalysis data at the anchorage position (05:25 UTC extraction). All reconstructed values are produced by the deterministic FishIntel pipeline. All reconstruction values are derived from verified FishIntel pipeline outputs.

PARAMETER	NARRATIVE DESCRIPTION	FISHINTEL RECONSTRUCTION	ASSESSMENT
Wind speed (sustained)	Beaufort F9–10 (47–55 kt)	27.0 kt	*Lower than narrative Beaufort estimate. Likely reflects separation between local 10 m wind and swell-driven sea-state severity. See analytical note below.*
Wind gusts	"gusting over 52 knots"	44.8 kt	*Broadly consistent with severe gust loading. ERA5 gust at 10 m is within ~15% of bridge-height observation. Height correction partially closes remaining gap.*
Wind direction	Southerly	179°	*Confirmed due south. Consistent*

PARAMETER	NARRATIVE DESCRIPTION	FISHINTEL RECONSTRUCTION	ASSESSMENT
			with lee shore exposure at anchorage.
Significant wave height	4–5 m swell	4.14 m	*Strong alignment with the narrative 4–5 m swell. Validates that ERA5 captures the same storm event.*
Atmospheric pressure	*Not stated*	`1004.4 hPa`	*Below-average pressure consistent with active low. FishIntel adds quantification absent from narrative.*
Air temperature	*Not stated*	`11.8 °C`	*Mild for winter; consistent with maritime air mass from the south.*
Dewpoint spread	*Not stated*	`3.2 °C`	*Indicator of precipitation / visibility conditions.*
Visibility classification	*Not stated*	`moderate`	*Derived from dewpoint spread.*
Wind vs tide	*Not stated*	`with`	*Interaction classification.*

PARAMETER	NARRATIVE DESCRIPTION	FISHINTEL RECONSTRUCTION	ASSESSMENT
Wind shear (10 m – 100 m)	***Not stated***	**12.7 kt**	*Significant shear. Relevant to high-windage vessel in ballast — wind loading at bridge height (~25–30 m) materially exceeds 10 m reference.*

27.0 kt SUSTAINED WIND (10 M)	44.8 kt GUST SPEED	4.14 m SIG. WAVE HEIGHT	1004.4 hPa PRESSURE MSL

ANALYTICAL NOTE — SUSTAINED WIND VS SEA-STATE SEVERITY

The narrative account describes conditions as "Beaufort force 9 to 10" (47–55 knots sustained). ERA5 reconstruction shows 10 m sustained wind of 27.0 knots (Beaufort force 6–7) at the anchorage position. These are not equivalent measurements and the difference is analytically significant.

ERA5 reports the 10-metre neutral-stability wind speed averaged over a ~31 km grid cell. The MAIB Beaufort classification is derived from sea-state observation at the vessel's position — a method that conflates locally generated wind waves with remotely generated swell. In this case, a 4+ metre swell was propagating into the anchorage from the open English Channel, generated by stronger winds over a long fetch offshore. The sea state at the anchorage therefore appeared consistent with Force 9–10 conditions, even though the local sustained wind at 10 m was materially lower.

This distinction matters operationally. The dominant forcing mechanisms for anchor drag in this case were heavy swell loading on a vessel in ballast (high windage area, large heaving motion in 4 m swell) combined with severe gust loading (45 kt at 10 m, likely exceeding 50 kt at bridge height). Sustained local Force 9–10 wind was not required to produce the observed outcome.

Unit Conversion Chain — Wind Speed

ERA5 source: **13.875** m/s → × (3600 / 1852) = **27.0 kt**

Unit Conversion Chain — Pressure

ERA5 source: **100438.8** Pa → ÷ 100 = **1004.4 hPa**

Unit Conversion Chain — Temperature

ERA5 source: **284.956** K → − 273.15 = **11.8 °C**

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Decision-Time Environment

The following chronology presents the sequence of events from the vessel's arrival at anchor through the grounding, with environmental context at each decision gate. Markers indicate points where the master had the opportunity to alter the outcome.

Day 1 — 1700 UTC

Vessel anchored in the bay

Port anchor deployed with 7 shackles of cable in 17 m charted depth, approximately 1 nm from the beach. Marine diesel delivered. No spare lube oil on board. Scope ratio ~9:1 at high water (21 m).

Day 3 — Evening

Decision: remain at anchor overnight

Master aware of forecast: southerly winds up to Beaufort force 8 (34–40 kt). Lube oil delivery expected but not completed. Night orders issued: engine readiness required at wind speeds >16–17 m/s (31–33 kt). Anchor drag risk discussed with bridge watchkeepers.

```
Forecast: S'ly F8 (34–40 kt) | Decision window: OPEN
```

Day 3 — 2100 UTC

Second officer takes bridge watch

Wind speed increasing. Engine placed at immediate readiness on second officer's orders.

```
Conditions escalating — wind increasing, gusts approaching 40 kt
```

Day 4 — 0400 UTC

Chief officer takes bridge watch

Southerly wind gusting over 52 knots. Swell 4–5 metres. Vessel yawing at anchor.

```
Conditions exceed forecast — 26.2 kt sustained, 43.7 kt gusts, 4.0 m
SWH | Decision window: NARROWING
```

Day 4 — 0425 UTC

Anchor starts to drag

Anchor watch alarm sounds on electronic chart system. SOG 1.4 kt to the north. Chief officer calls master, alerts engine room, announces to crew.

```
HDG 186° | SOG 0.4 kt | COG 032° | 26.2 kt sustained, 43.7 kt gusts
```

Day 4 — 0438 UTC

Heaving anchor

Bosun begins heaving the anchor. Master uses engine and rudder to take weight off the cable.

```
HDG 208° | SOG 1.5 kt | COG 319°
```

Day 4 — ~0500 UTC

Anchor aweigh — fouled by chain

Anchor broke the surface fouled by a substantial length of chain across its flukes. Master stopped manoeuvring, concerned about damage to anchor, windlass, or propeller. Vessel fell beam-to the wind on a westerly heading.

```
HDG 216° | SOG 0.6 kt | COG 289° | Decision window: CLOSING
```

Day 4 — 0510 UTC

Final manoeuvring attempt

Vessel approximately 6 cables off the beach, making over 2 kt leeway to the north. Full port rudder and full ahead applied. Master assessed vessel would not clear the headland 0.5 nm to the west.

```
HDG 274° | SOG 2.1 kt | COG 334° | 27.0 kt sustained, 44.8 kt gusts
```

Day 4 — 0515 UTC

Engine stopped

Master assessed grounding was inevitable and stopped engine to protect the propeller for future refloating. Ordered port anchor let go — anchor did not hold.

```
HDG 254° | SOG 1.1 kt | COG 342°
```

Day 4 — 0525 UTC

Grounding

Vessel grounded on the beach. Settled with 5° port list. Stern on sand, bow across rocky ledge. Master informed port and vessel operator.

```
HDG 240° | SOG 0.8 kt | COG 333° | Decision window: CLOSED
```

Day 4 — 1415 UTC

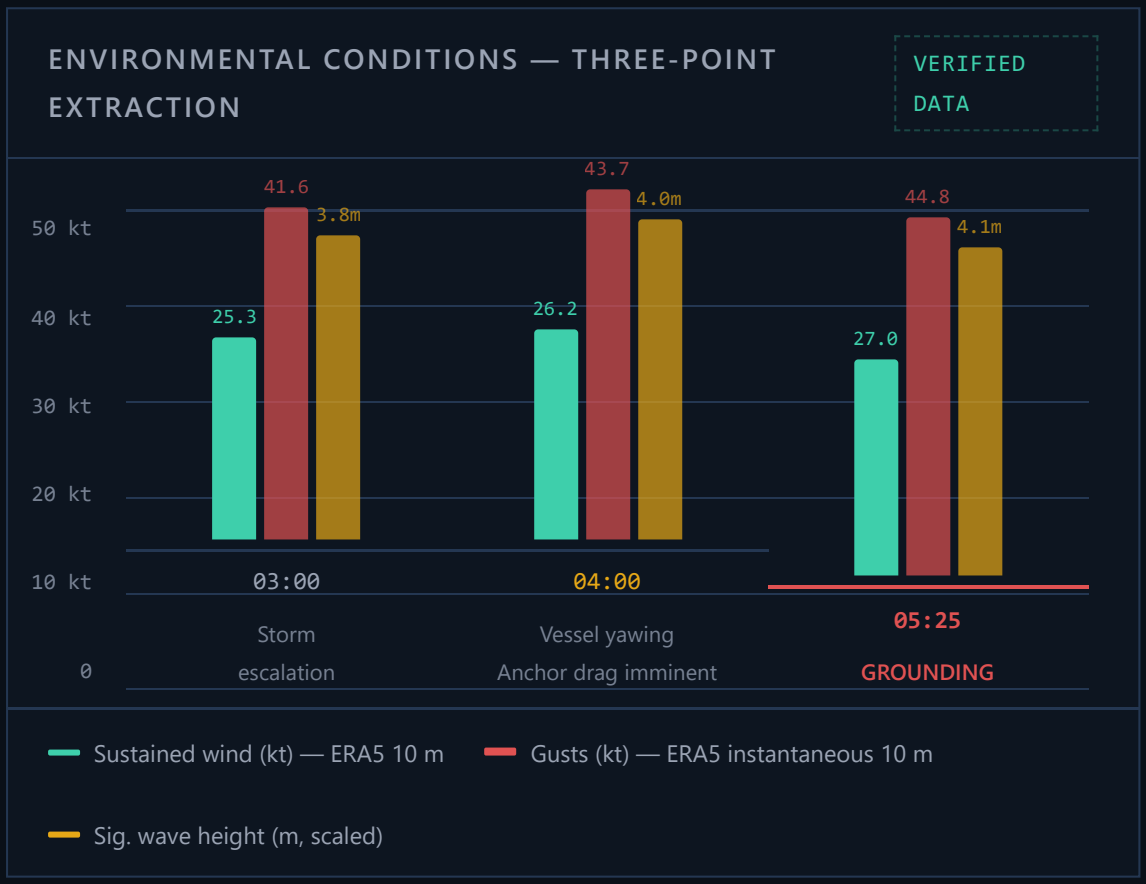
Refloated

Vessel refloated by local harbour tugs and own main engine as sea and swell moderated. Anchored 1 nm south of the headland by starboard anchor.

07

Environmental Escalation — Verified Profile

The following graphic presents the verified three-point environmental escalation profile across the incident window. Each data point is derived from the FishIntel reconstruction pipeline at the coordinates and timestamp shown.



INTERPRETATION

The three-point extraction shows a slowly escalating environmental regime, not a sudden onset event. All three extraction points classify as HIGH risk. The conditions that caused the anchor to drag at 04:25 were already present at 03:00 — more than 90 minutes earlier. Across the extraction window, sustained wind increased by 1.7 knots and gusts by 3.2 knots.

Gusts consistently exceeded the master's night-order threshold of 31–33 knots by a margin of 10–14 knots across the entire window. The forecast available to the master (southerly F8, 34–40 kt) understated the gust intensity that materialised.

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Decision Window Assessment (Preliminary)

FishIntel's Time-to-Irreversibility (T_{irr}) framework identifies when an operational situation becomes irreversible — the point beyond which no reasonable intervention by the vessel's crew can change the outcome. This preliminary assessment uses the three-point environmental extraction to establish the regime classification across the incident window.

The vessel was in a HIGH-risk environmental regime at all three extraction points (03:00, 04:00, 05:25 UTC). The hazardous state pre-dated the anchor drag event by at least 90 minutes.

Decision Window Findings (Preliminary)

RISK REGIME	HIGH at all verified extraction points (03:00, 04:00, 05:25 UTC)
INDICATIVE WARNING WINDOW	Hazardous anchoring regime established at least 90 minutes before anchor drag
INDICATIVE LOSS OF REVERSIBILITY	Operational recovery appears to have narrowed rapidly after anchor drag began at 04:25 UTC, with the vessel beam-to the wind and significant leeway developing by approximately 05:10
CRITICAL DECISION GATE	Evening of Day 3 — decision to remain at anchor on a lee shore with storm forecast

AVAILABLE ALTERNATIVE	Proceed to sea before storm escalation, while the decision window remained open

Precise irreversibility timing and regime-transition thresholds are reserved for full T_{irr} computation with vessel-specific CaseContext parameters. The findings above are derived from the verified environmental reconstruction and the incident chronology.

The T_{irr} framework classifies operational situations into three regimes:

REGIME	CLASSIFICATION	MEANING
Regime 1	**Reversible**	*Conditions permit effective intervention. The decision window is open. Corrective action remains available and viable.*
Regime 2	**Conditionally reversible**	*Intervention is possible but constrained. The window is narrowing. Success depends on immediate action and favourable conditions.*
Regime 3	**Irreversible**	*No reasonable intervention can alter the outcome. The situation has passed T_{irr}. The operational decision that could have changed the outcome had to be taken earlier.*

Day 3 evening — Decision to remain at anchor0400 — Conditions exceed forecast0510 — Grounding inevitable

Based on the narrative evidence, this case presents a three-phase decision window. On the evening before the grounding, the decision to remain at anchor represented a point where the situation was reversible — the vessel could have proceeded to sea despite its operational constraints. By 0400, when conditions exceeded the forecast available to the master, the window was

narrowing but action remained possible. By approximately 0510, with the vessel beam-to the wind making over 2 knots leeway at 6 cables from the beach, the outcome had become inevitable regardless of crew action.

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Risk Analysis

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Risk Level:

HIGH

Decision classification:

DANGEROUS

The FishIntel risk assessment integrates multiple environmental drivers to produce a single quantified score. The drivers for this case include:

RISK DRIVER	VALUE	CONTRIBUTION
Sustained wind	27.0 kt	*Primary forcing. Dominant driver of anchor drag and leeway.*
Gust speed	44.8 kt	*Peak loading on anchor cable. Gust penalty applied.*
Significant wave height	4.1 m	*Swell-induced heaving increases anchor cable loading cycles.*
Wind vs tide	with	*Tide multiplier applied to base risk.*
Visibility	moderate	*Situational awareness factor.*
Wind shear	12.7 kt	*Relevant to high-windage vessel in ballast condition.*

A vessel in ballast presents a substantially larger windage area than a laden vessel. The combination of sustained storm-force winds, heavy swell, and lee shore exposure at the anchorage position created conditions under which anchor drag was a foreseeable outcome for a vessel of this size and condition.

◆ FORESEEABILITY TEST

This report assesses foreseeability by applying the following test to the reconstructed environmental evidence:

- The environmental data required to assess the risk existed and was accessible at the time of the decision (ERA5 reanalysis confirms this retrospectively; forecast data confirms it prospectively).
- The data was sufficient to identify that conditions at the anchorage position would exceed the operational envelope for a vessel of this type, size, and ballast condition.
- A reasonable operator with access to this data would have identified the risk of anchor drag under the forecast conditions and considered proceeding to sea or relocating to a more sheltered anchorage.
- The environmental progression from forecast conditions to actual conditions was continuous, not discontinuous — there was no sudden, unforeseeable step-change that made the outcome unavoidable.

This report does not attribute fault or negligence to any individual. The foreseeability test evaluates the environmental evidence, not the decisions of specific persons. Its purpose is to establish whether the conditions that caused the incident were objectively knowable before the outcome occurred.

Key Analytical Finding

KEY FINDING

Environmental reconstruction separates local wind forcing from remotely generated swell, revealing that the conventional Beaufort force 9–10 classification in the narrative account conflated two distinct physical mechanisms. ERA5 reanalysis shows 10 m sustained winds of 25.3–27.0 knots (Beaufort force 6–7) at the anchorage position, with gusts of 41.6–44.8 knots and significant wave height of 3.8–4.1 metres. The dominant forcing for anchor drag was heavy swell loading — generated by open-ocean winds over a long English Channel fetch — combined with severe gust intensity on a vessel in ballast with high windage. The hazardous anchoring regime was already established by 03:00 UTC, more than 90 minutes before the anchor began to drag.

PIPELINE CLASSIFICATION

CRITICAL: Environmental reconstruction shows the vessel was in a maximum-risk (100%) anchoring regime across the entire incident window (03:00–05:25 UTC). The dominant forcing was heavy swell (3.8–4.1 m SWH) propagating into the anchorage from the open English Channel, combined with severe gust loading (41.6–44.8 kt). Sustained 10 m wind of 25–27 kt (Beaufort force 6–7) was materially lower than the narrative Force 9–10 description, which reflects a sea-state-based Beaufort estimate conflating remote swell with local wind. The hazardous regime pre-dated the anchor drag event by at least 90 minutes.

This finding is derived from the quantified environmental reconstruction, not from the narrative account. It represents what was objectively knowable at the time the critical operational decision was made, based on environmental data that existed and was accessible.

This report does not attribute fault or negligence to any individual. The master was aware of the forecast, implemented precautionary measures, and took prompt action when the anchor began to drag. The purpose of the reconstruction is to quantify the environmental conditions and the constraints under which operational decisions were made.

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Evidence Integrity

This report is a rendering of outputs produced by the FishIntel deterministic reconstruction pipeline. All environmental values presented are traceable to their source data through an auditable processing chain.

PIPELINE VERIFICATION

DATA SOURCE	ECMWF ERA5 Atmospheric Reanalysis
PROCESSOR VERSION	5.3.0
RUNNER VERSION	2.2.0
GATE CONFIG HASH	989aadf7b9b1a4acef8606e162015337e75a8ce5fcb4edfb1e0d79f4d59513aa
RECONSTRUCTION TIMESTAMP	2026-03-30T22:10:21Z
RAW FORMAT	era5_raw_v1
CASE POSITION	50.1456°N, 5.0667°W
CASE TIMESTAMP	2018-12-18T05:25:00

The pipeline operates as follows: ERA5 NetCDF data is extracted at the incident coordinates and time window by the ERA5 processor, producing raw values in source-native units (m/s, Pa, K, m). These raw values pass through FIG_UNIT_GATE, which performs Decimal-arithmetic unit conversions and plausibility checks before any value enters the reconstruction engine. The gate configuration hash cryptographically identifies the exact conversion factors, plausibility bounds, and variable registry active at the time of processing.

No manual adjustments, interpolations, or subjective corrections are applied at any stage. The reconstruction is deterministic: the same inputs will always produce the same outputs.

Methodology

FishIntel Global reconstructs the environmental conditions surrounding maritime incidents using ECMWF ERA5 atmospheric reanalysis data — the same dataset relied upon by meteorological agencies, climate researchers, and aviation safety authorities worldwide.

ERA5 provides hourly estimates of atmospheric, land-surface, and ocean-wave variables on a global grid at approximately 31 km horizontal resolution. It assimilates millions of observations per cycle and is continuously quality-controlled by the European Centre for Medium-Range Weather Forecasts. ERA5 data is freely available and independently verifiable by any party.

The FishIntel reconstruction pipeline extracts environmental variables at the precise coordinates and time window of each incident. All values pass through a unit-gated validation layer that enforces explicit conversion factors and plausibility bounds using Decimal arithmetic — preventing the silent unit errors that are a known source of analytical failure in environmental data processing. The pipeline is deterministic: identical inputs always produce identical outputs.

This reconstruction operates within defined boundaries. ERA5 is a gridded reanalysis product, not a point measurement. It does not capture localised effects smaller than its grid resolution, such as harbour-scale wind funnelling or shallow-water wave refraction. Where a case involves such effects, the reconstruction states this limitation explicitly. The system is designed to produce defensible evidence, not to overstate its precision.

FishIntel's role is complementary to national maritime accident investigation. Investigation bodies establish causes and circumstances. FishIntel quantifies the environmental context in which operational decisions were made, using

data sources that existed at the time of those decisions. The focus is on what was knowable, not on who is to blame.

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Closing Statement

Maritime incidents are frequently described after the fact using qualitative or semi-quantitative environmental descriptions: "strong winds," "heavy swell," "poor visibility." These descriptions are valuable for narrative understanding but do not quantify the decision-time environment with the precision required by insurers, regulators, and legal professionals.

Environmental reconstruction bridges this gap. By reconstructing the conditions that existed at the time a critical decision was made — using data sources that were available or forecastable at that time — it becomes possible to evaluate what was objectively knowable before the outcome occurred.

This does not mean that every incident was preventable. It means that the environmental context of every maritime decision can now be quantified, compared against published forecasts, and assessed for foreseeability. The data exists. The question is whether it was used.

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